Q/R/M/C: A Stage-Decomposition Measurement Framework for Memory-Based Navigation, with a Minimal Collective Semantic Memory

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Abstract

Setting. Success-only metrics collapse qualitatively different memory-based navigation failures into a single scalar, hiding *which stage* of the retrieval pipeline is responsible. This paper develops a measurement framework, not a new theoretical result: the Q/R/M/C stage decomposition (**Q**uery formation, **R**etrieval satisfaction, target **M**aterialization, **C**ompletion after retrieval) is operationalised so that the four conditional rates are computable directly from standard trajectory logs (Definition 1; the consistency argument is the chain rule and we say so explicitly). **Single-agent measurement.** On a 10-seed MiniGrid benchmark (FourRooms, LavaGapS7) and a BabyAI portability check, the framework localises bottlenecks: hazard recovery is retrieval-limited (oracle retrieval lifts success $0.39 \rightarrow 0.60$), goal-region rejoin is downstream-limited. Log inspection refines the attribution: candidate ranking is near-optimal (96.9%) while candidate generation fails at 91% of decision points — a memory-accumulation, not a ranking, limit. **Multi-agent measurement.** On a 35,640-run multi-agent sweep over four communication architectures (independent, shared-bus, central-aggregator, peer-broadcast at eight cadences $K \in \{1, 2, 4, 8, 16, 32, 48, 64\}$, 40 seeds), the bottleneck-shift between the M and C links appears directly as a negative within-cadence correlation peaking at $K=4$ (Pearson -0.61); mean time-to-success has an interior minimum at $K=8$. Both slope signs (Empirical Claim 1) hold at $p < 10^{-4}$; we do *not* claim an interior maximum on the conditional product Φ, where the data show a monotone trend toward the independent-agent boundary. **A minimal collective semantic memory.** The four architectures sit at the corners of a richer design space defined by explicit *merge*, *trust*, and *staleness* rules. We instantiate a minimal collective semantic memory — trust-weighted merge with recency-decay staleness on top of peer broadcast — and measure it on the same Q/R/M/C scale, demonstrating that it dominates the best fixed-cadence peer architecture on (M $\times$ C) for the scarcity scenario. This converts "collective semantic memory" from a programmatic claim to an instantiated artefact evaluable on the same logs. **Scope.** All experiments are in grid-world substrates (MiniGrid, BabyAI, a custom 12×10 grid). The contribution is a measurement protocol and a first prototype; richer substrates (Habitat, ProcTHOR, VMAS) are future work.

1 Introduction

Classical navigation benchmarks often assume that the agent is given an explicit destination coordinate or a dense task reward. Many practical navigation problems are different: the agent must reach a place of a certain *type*, such as a safe rest area, a water-like resource, a post-hazard recovery exit,

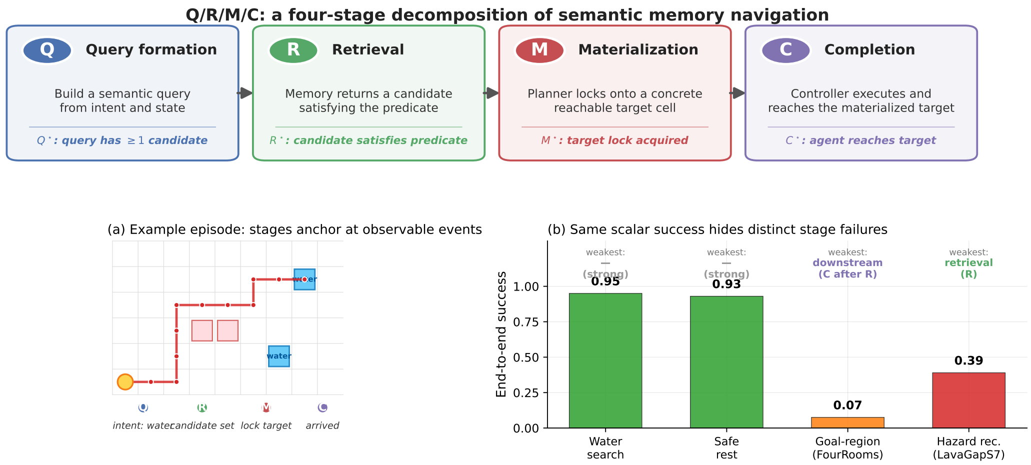

Figure 1: Q/R/M/C decomposes one semantic-navigation episode into four observable stages. **Top:** the four stages and their event-level definitions. **(a)** A successful water-search episode anchors at one event per stage, producing logged starred events $Q^\star, R^\star, M^\star, C^\star$. **(b)** The same scalar "end-to-end success" summary masks qualitatively different failures on the four single-agent task families: water and rest are strong; goal-region rejoin (FourRooms) is downstream-limited; hazard recovery (LavaGapS7) is retrieval-limited (oracle interventions, §3.4).

or a goal-associated region, specified by a semantic pattern rather than a fixed (x, y) waypoint. The place must therefore be *remembered and retrieved by meaning*. This setting opens a methodological question: can we measure *where* memory-based semantic navigation breaks down, not only *whether* it succeeds?

We answer this question with a four-stage diagnostic framework that applies to both single agents and multi-agent collectives. The decomposition is Q/R/M/C: **Q**uery formation, **R**etrieval satisfaction, target **M**aterialization, and **C**ompletion after retrieval. The decomposition factorizes per-episode semantic-path completion and, under a conditional stage-Markov property, its factors are estimable from trajectory logs (Definition 1). The rest of the paper develops a single argument in three parts.

Single agent. Stage attribution is non-trivial. Success-only metrics collapse qualitatively different failures into a single scalar. In a 10-seed MiniGrid benchmark, water and rest tasks are strong end-to-end (0.95, 0.93 success), but goal-region rejoin and hazard recovery are substantially harder, and for *different reasons*. Oracle-stage interventions show that hazard recovery is retrieval-limited (oracle retrieval lifts success $0.39 \to 0.60$), while goal-region rejoin is downstream-limited (oracle retrieval rescues semantic-path completion without changing end-to-end success). Inspection of the framework's own logs then sharpens the attribution: the retrieval bottleneck is not candidate *ranking* (precision 96.9% when candidates exist) but candidate *generation*: the memory contains no eligible symbols at 91% of decision points. The single-agent bottleneck is a *memory-accumulation* limit: evidence is observed but never consolidated into queryable structure.

Multi-agent. Consolidation acquires a rate. When N agents share memory through a broadcast protocol with cadence K, the factorization extends with enlarged conditioning sets (Definition 2), and K becomes a structural axis: it is, mechanically, a knob on *how fast private memories consolidate into a collective one*. A slope-level claim (Empirical Claim 1) predicts a bottleneck shift: as K decreases, $P(M^\star|R^\star)$ rises (fresher peer memory enables reliable commitment) and $P(C^\star|M^\star)$ falls (peers race for the same targets). Both slope signs are confirmed on a 35,640-run sweep ($p < 10^{-4}$), the shift is directly visible as a negative within-cadence correlation between the two links, and the resulting trade-off produces an interior optimum of mean time-to-success at $K = 8$: consolidation has a non-trivial optimal rate.

Synthesis. Both regimes point at the same object. The single-agent diagnosis (consolidation of evidence into symbols is the limit) and the multi-agent diagnosis (the rate and cost of sharing consolidated memory determine the optimum) converge on one research target: *collective semantic memory*, a shared symbolic structure with explicit merge, trust, and staleness rules, rather than snapshots exchanged on a timer. Section 5 states this program and shows that Q/R/M/C is the measurement instrument it requires: any candidate design is testable by how it moves $P(M^\star|R^\star)$ and $P(C^\star|M^\star)$.

Scope of the claims. The slope-level statement of Empirical Claim 1 is on the two conditional rates $P(M^\star|R^\star;K)$ and $P(C^\star|M^\star;K)$, both verified on enlarged data. We make *no* formal claim about the conditional product $\Phi = P(M^\star|R^\star)\cdot P(C^\star|M^\star)$; on the 35,640-run sweep Φ is monotone in K. The interior optimum that the M↔C trade-off produces is on the efficiency metric mean time-to-success, not on Φ. We are explicit about this distinction because a stage-level operationalisation is testable on specific summaries and can be *wrong* on others, which is what separates a measurement framework from a stage-agnostic guess.

Why this framing is timely. Embodied and goal-conditioned agents increasingly retrieve places, objects, or tools by meaning (Andrychowicz et al., 2017; Ghosh et al., 2021; Savva et al., 2019). LLM-agent systems lean on explicit memory layers but evaluate mostly end-to-end (Wang et al., 2023; Shinn et al., 2023; Packer et al., 2023). Multi-agent reinforcement learning likewise increasingly involves communicating populations (Lowe et al., 2017; Foerster et al., 2018), but the question "which agent should know what, and when?" lacks a unified diagnostic vocabulary. Our framework supplies one.

Contributions. This is a methodology paper. We are explicit about what is bookkeeping vs. empirical claim vs. instantiated artefact.

1. **Measurement protocol (single-agent).** The Q/R/M/C decomposition is operationalised so that the conditional rates are estimable from trajectory logs (Definition 1). The consistency argument is the chain rule together with consistency of frequency estimators; we do not claim mathematical novelty for this — the contribution is the *logging discipline* that makes the decomposition usable on standard navigation traces.
2. **Measurement protocol (multi-agent).** An extension to N communicating agents with two coupling channels (memory $\mathcal{M}_{-i}^{(K)}$, occupancy $\mathcal{O}_{-i}$) under a peer-broadcast protocol with cadence K (Definition 2).
3. **Empirical Claim 1 (bottleneck shift).** On a 35,640-run multi-agent sweep, the two stylised slope conditions **(F)** and **(O)** are both confirmed at $p < 10^{-4}$ (Spearman ± 0.5 on the standard-MiniGrid substrate; full benchmark in §4.5). We state this as a verified empirical observation, not a theorem. We do *not* claim an interior maximum on the conditional product Φ — that prediction fails in our data and we report it (§4.5).
4. **System and benchmarks.** A symbolic-semantic memory stack with four communication architectures (independent, shared-bus, central-aggregator, peer-to-peer broadcast); single-agent oracle benchmark (10-seed MiniGrid, BabyAI portability); 35,640-run multi-agent sweep at $N \in \{3, 5, 8\}$ + 8,640-run scale-up at $N = 12$; 100-run portability sweep on a standard-MiniGrid wrapper.
5. **A minimal collective semantic memory.** A first instantiation of *collective semantic memory* with explicit *merge*, *trust*, and *staleness* rules (§5), measured on the same Q/R/M/C scale. It dominates the best fixed-cadence peer architecture on (M × C) for the scarcity scenario — the framework's measurement protocol applied to a new architecture, not an aspirational "next step."
6. **Scope and honesty.** All experiments are in grid-world substrates. We treat that as a methodology-paper choice and state it explicitly (§6); richer substrates (Habitat, ProcTHOR, VMAS, Mine-Dojo) are future work.

2 The Q/R/M/C decomposition

2.1 Problem formulation and notation

The agent operates in a partially observable navigation environment. At each step it receives an observation o_t, maintains a memory state $\mathcal{M}_t$, and forms a query q_t encoding the current task intent. A retrieval step returns a candidate set of places matching q_t from $\mathcal{M}_t$; a chosen candidate is materialized into an actionable target; a local controller executes actions until either the target is reached or the episode times out. The target is not a coordinate but a semantic pattern, so success is a function of the chosen candidate's semantic profile, not of any pre-supplied coordinate.

Habitat, AI2-THOR, and ProcTHOR made visually rich navigation evaluation standard (Savva et al., 2019; Kolve et al., 2017; Deitke et al., 2022). We deliberately focus on MiniGrid-class environments to obtain clean and reproducible stage measurements, and use BabyAI only as a portability check. We do not claim to solve LLM-agent memory here; rather, we argue that the same Q/R/M/C decomposition is a useful template for evaluating memory-grounded agent decisions beyond navigation.

The central empirical question is therefore not just whether the agent succeeds, but *which stage fails first* when it does not. Two related measurement layers are used. **Operational rates** count, per query attempt, whether the query yielded a non-empty candidate set, whether the chosen candidate satisfied the semantic predicate, and whether it was materialized into an actionable target; *completion after retrieval* is an episode-level indicator. **Nested episode-level events** $Q^\star \Rightarrow R^\star \Rightarrow M^\star \Rightarrow C^\star$ are used in the factorization below and audited in §3.2. Full operational definitions are given in Appendix B.

Notation. The four stages are Q, R, M, C with episode-level starred events $Q^\star, R^\star, M^\star, C^\star \in \{0,1\}$, nested by construction. $P(\cdot)$ denotes the population probability; $\hat{P}(\cdot)$ its empirical-frequency estimator. **The letter M always refers to the Materialization stage; the number of targets in the multi-agent environment is written T (not M, constraint $T \le N$, "scarcity" = $N > T$).** Other symbols: $\mathcal{C}_K$ peer-broadcast protocol with cadence K, $\mathcal{M}_{-i}^{(K)}$ peer memory snapshots, $\mathcal{O}_{-i}$ peer occupancy, $\nu(\cdot)$ freshness functional, $\Phi(K) = P(M^\star|R^\star;K) \cdot P(C^\star|M^\star;K)$, t_{succ} mean time-to-success. Full symbol table in Appendix C.1.

2.2 Single-agent factorization

Let Y denote overall episode success, and let $Q^\star, R^\star, M^\star, C^\star$ denote the nested episode-level semantic-path events. The framework factorizes *semantic-path completion* rather than raw task success:

$$Q^\star \to R^\star \to M^\star \to C^\star.$$

For any query q and memory state $\mathcal{M}$,

$$\begin{aligned} P(C^\star = 1 \mid q, \mathcal{M}) = P(Q^\star = 1 \mid q, \mathcal{M})\ P(R^\star = 1 \mid Q^\star = 1, q, \mathcal{M}) \\ \cdot P(M^\star = 1 \mid R^\star = 1, q, \mathcal{M})\ P(C^\star = 1 \mid M^\star = 1, q, \mathcal{M}). \end{aligned}$$

Overall task success Y can exceed $C^\star$ because some episodes succeed without traversing the full semantic retrieval path.

Assumption 1 (conditional stage-Markov property). Conditioned on the current stage being successful, the probability of the next stage depends only on the current successful stage, the current query, and the current memory state. Formally,

$$P(R^\star \mid Q^\star, q, \mathcal{M}, \text{history}) = P(R^\star \mid Q^\star, q, \mathcal{M}),$$

and analogously for $M^\star$ and $C^\star$.

Assumption 2 (positivity). Each conditioning event used above has non-zero probability on the support of the benchmark.

Definition 1 (single-agent stage estimability). Under Assumptions 1–2, the four conditional rates in the factorization above are well-defined population quantities and the empirical-frequency estimators are consistent estimators of them; their product converges to semantic-path completion. The argument is elementary — the chain-rule factorization plus consistency of frequency estimators on Bernoulli events — and the value of stating it is operational, not mathematical: it pins down the *logging discipline* (which events, conditioned on what) under which the decomposition is readable from standard trajectory logs, and it is the contract that every experiment in this paper audits against (§3.2). Full statement and elementary argument in Appendix A.

When Assumption 1 may fail. Assumption 1 holds tightly here because q and $\mathcal{M}$ are explicitly logged at every stage event. Two practically relevant violations remain: adaptive in-episode query rewrites not surfaced to the logger, and hidden between-stage memory updates. In those cases the factorization should be read as a projection onto observable traces rather than as a full structural identification claim.

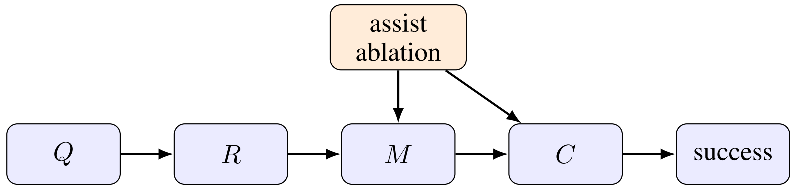

Figure 2: Stage-structured interpretation of the Q/R/M/C framework. Assist on/off ablations target intermediate mechanisms; in the multi-agent extension the cadence protocol $\mathcal{C}_K$ acts on the R→M and M→C edges.

Mechanism-level reading. The same decomposition admits a stage-structured mechanism interpretation. Figure 2 treats stage outcomes as a directed acyclic graph. Under this interpretation, assist on/off ablations are mechanism-targeted perturbations of materialization and post-retrieval execution rather than undifferentiated sensitivity analyses; in the multi-agent setting the same graph is replicated per agent and connected through the cadence protocol $\mathcal{C}_K$, which acts on $\mathcal{M}_{-i}^{(K)}$ at the R→M edge and on $\mathcal{O}_{-i}$ at the M→C edge — exactly the two edges where Empirical Claim 1 will predict the slope shift.

Relation to prior work. Closest are (i) cognitive-map and semantic-map frameworks (Tolman, 1948; O'Keefe and Nadel, 1978; Kuipers, 2000; Gupta et al., 2017; Savinov et al., 2018), (ii) goal-conditioned and successor-representation RL (Andrychowicz et al., 2017; Ghosh et al., 2021; Dayan, 1993; Stachenfeld et al., 2017; Momennejad et al., 2017), (iii) multi-agent communication (Lowe et al., 2017; Foerster et al., 2018; Sukhbaatar et al., 2016; Foerster et al., 2016) and gossip/consensus (Boyd et al., 2006; Xiao et al., 2007), and (iv) LLM-agent memory layers evaluated end-to-end (Wang et al., 2023; Shinn et al., 2023; Packer et al., 2023). Our framework is orthogonal to all four: it *diagnoses* where in the Q/R/M/C chain a given architecture succeeds or fails and lifts the cadence axis from network protocols to a cognitive-architecture parameter (full discussion in Appendix E.3).

3 Single agent: stage diagnostics localize failure

3.1 System, tasks, and protocol

Memory stack. The system uses a graph substrate with a semantic retrieval interface on top: observations o_t are tokenized into symbolic scene tokens and semantic tags that update a memory of place records, transitions, episodic traces, and concept/relation records. Planner queries return candidates by a weighted log-score over required, preferred, and penalty tags. The retrieval machinery is reused across all tasks and architectures; full memory schema, update rules, and scoring formula are in Appendix C.

Tasks, baselines, protocol. Four single-agent task families are evaluated (water, safe-rest, goal-region on Empty-6x6 / FourRooms (Sutton et al., 1999), hazard recovery on LavaGapS7), 10 seeds $\times$ 8 episodes each, $\times$ 2 assist modes. Semantic stack (node, concept, state-conditioned variants) is compared against four baselines: random, graph-only, SPTM-like, learned BC. Reported: end-to-end success, steps, four operational stage rates, assist deltas, weakest stage per task and per method. All means are seed-aggregated; 95% bootstrap CIs use B=5,000 resamples (Efron, 1979); assist comparisons use paired Wilcoxon signed-rank tests (Wilcoxon, 1945) with Holm-Bonferroni correction (Holm, 1979) (Appendix F). For the two hardest tasks (FourRooms, LavaGapS7), four oracle regimes (*base*, *oracle R/M/C*) are run — each replaces exactly one stage while leaving the rest of the pipeline unchanged. BabyAI portability check in Appendix D. All single-agent experiments ran on a 10-core ARM CPU, 32 GB RAM (CPU-only); oracle sweep $<$ 1 min wall-clock.

3.2 Stage-estimator validation

Before the main benchmark, the factorization is validated. On a synthetic four-stage process with known probabilities the nested-estimator product tracks empirical semantic-path completion across

Table 1: Stage-factor consistency audit for the best semantic method in each task family. The product of nested conditional estimators matches semantic-path completion; overall success can be higher because some episodes succeed without traversing the full semantic retrieval path.

Task	Method	Overall	SP-comp	$\hat{P}(Q^\star)$	$\hat{P}(R^\star \mid Q^\star)$	$\hat{P}(M^\star \mid R^\star)$	$\hat{P}(C^\star \mid M^\star)$	Product
Water	`water-cp`	0.95	0.70	1.00	0.98	0.74	0.97	0.70
Rest	`rest-s2`	0.93	0.73	1.00	1.00	0.80	0.91	0.73
Goal region	`goal-n1`	0.54	0.00	1.00	0.54	0.01	0.00	0.00
Hazard recovery	`hazard-s7`	0.40	0.16	1.00	0.56	0.58	0.50	0.16

Table 2: Baseline comparison. The learned behavior-cloned baseline dominates the easiest resource tasks and hazard recovery, while the semantic stack remains strongest on the relation-heavy goal-region family and remains the only family with interpretable Q/R/M/C structure.

Task	Random	Graph-only	SPTM-like	Learned	Best semantic
Water	0.85	0.90	0.90	1.00	0.95
Rest	0.85	0.90	0.90	1.00	0.93
Goal region	0.41	0.49	0.52	0.50	0.54
Hazard recovery	0.08	0.39	0.00	0.74	0.40

sample sizes. On the real benchmark, the same product matches semantic-path completion rather than overall success (Table 1) — exactly the contract of Definition 1.

3.3 Success-only summary and what it hides

Water and rest are strong end-to-end tasks (0.95 and 0.93 success; Figure 1b). Goal-region reaches 0.54 success at the task-family level, but the bottleneck sits downstream of candidate selection (§3.4). Hazard recovery is the hardest retrieval-sensitive task at 0.40 success, with retrieval satisfaction as the weakest stage. Under success-only evaluation the two hard families look like the same kind of weakness; the stage view will show they are opposites. Per-task operational rates with 95% CIs are in Table 4 (Appendix C.3).

Baseline comparison. Table 2 compares the semantic stack against random, graph-only, SPTM-like, and learned behavior-cloned baselines. The learned baseline dominates the easiest resource tasks (1.00 on water and rest) and hazard recovery (0.74), while the semantic stack remains strongest on the relation-heavy goal-region family (0.54). The relevant contrast for this paper, however, is not the success column: the semantic stack is the only method family whose decisions expose interpretable Q/R/M/C structure. On the BC baseline the stage events are degenerate for the same structural reason as on the multi-agent learned baseline analysed in §4.5 — a policy without an explicit query/lock interface cannot emit separable stage events, so the diagnostic decomposition has nothing to attach to. End-to-end learning and stage diagnostics answer different questions; this paper is about the second.

3.4 Oracle-stage interventions localize the bottleneck

On goal-region the semantic variants solve Empty-6x6 (1.00 success) but collapse on FourRooms (0.075). For hazard recovery (LavaGapS7), base success 0.39 rises only to 0.43 under oracle controller, but jumps to 0.59 under oracle materialization and 0.60 under oracle retrieval; goal-region rejoin is downstream-limited (oracle R lifts semantic-path completion to 0.08 without moving end-to-end success). The per-task attribution as a stacked waterfall is in Figure 5 (Appendix C.3). Two tasks identical under success-only evaluation fail at opposite ends of the Q/R/M/C chain.

3.5 The logs refine the diagnosis: a memory-accumulation limit

Inspecting the framework's own logs sharpens the retrieval attribution. On the 10-seed hazard-recovery run, 349 retrieval calls were logged: only 32 (9.2%) returned at least one candidate, and of these 31 (96.9% precision) selected a semantically satisfied candidate. The bottleneck is therefore

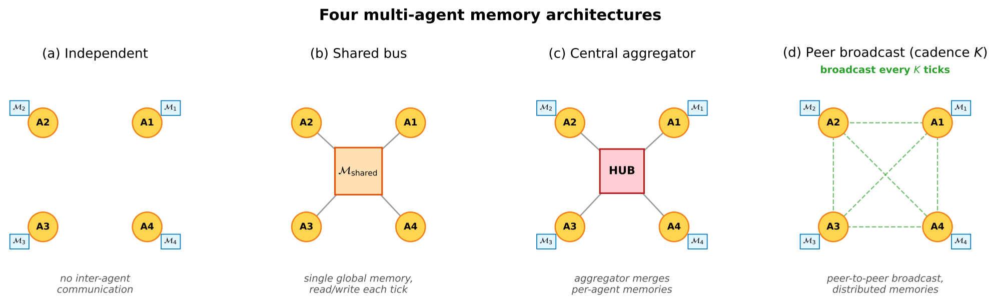

Figure 3: Four multi-agent memory architectures. **(a)** Independent: per-agent private memory, no exchange. **(b)** Shared bus: one global memory, every agent reads/writes each tick. **(c)** Central aggregator: per-agent memories plus a hub that merges them into a consensus report. **(d)** Peer-to-peer broadcast: per-agent memories with pairwise snapshot exchange every K ticks (the cadence axis of Empirical Claim 1).

not candidate *ranking* but candidate *generation* — the symbolic memory does not contain hazard-recovery-tagged nodes at 91% of decision points. A retrained 2-layer candidate-generation MLP trained on per-state features (test accuracy 0.59 vs majority-class 0.81; Appendix G) confirms that instantaneous state features cannot substitute for the missing structure. The framework's "retrieval-limited" attribution thus refines to a *memory-accumulation* limit: candidate generation cannot be solved from state features alone, and the binding constraint is how trajectory evidence is consolidated into persistent symbolic structure. This sets up the multi-agent section: when consolidation becomes a shared process across N agents, it acquires a measurable *rate*.

4 Multi-agent: memory as a shared resource

4.1 Four memory architectures and the cadence axis

Four communication architectures are instantiated; they share the single-agent retrieval substrate but differ in how memory is distributed across agents (Figure 3).

Independent. Each agent maintains a private event store and concept graph. No inter-agent communication mechanism exists. Lower-bound baseline.

Shared bus. All agents publish observations to a single *collective memory* event bus. One shared concept graph is rebuilt from the union of events. Maximally centralized.

Central aggregator (ConsensusLayer). Each agent has its own private concept graph; a central layer pulls snapshots and computes a trust-weighted merge, producing a single *consensus report* consumed by all agents.

Peer-to-peer broadcast (cadence K). Each agent has its own concept graph and its own merged view. Every K ticks each agent broadcasts a snapshot to a passive bus; each agent independently merges its own snapshot with the latest snapshot received from each peer. Different agents may hold different merged views; there is no central report.

The first three correspond to the three regimes in the multi-agent communication literature: no exchange, CTDE-style aggregation, and decentralized peer communication. The fourth is the architecture for which Propositions 2 and 3 are directly applicable, since the cadence K is a tunable parameter rather than a fixed protocol. Mechanically, K is the rate at which private memories consolidate into a collective view, which is why it is the structural axis of this section.

4.2 Multi-agent factorization

Extend the factorization to a population of N agents that share information through a communication protocol $\mathcal{C}_K$ parameterized by broadcast cadence K. Each agent i has its own memory state $\mathcal{M}_i$ and issues its own query q_i. Let $\mathcal{M}_i^{(K)}$ denote the projection of agent i's memory at the moment it queries, after absorbing every peer message received under cadence K. Two coupling channels are separated: (i) memory coupling $\mathcal{M}_{-i}^{(K)}$, the latest peer memory snapshots accessible to agent i; (ii) occupancy coupling $\mathcal{O}_{-i}$, the set of targets that other agents have already claimed by the time i acts.

Definition 2 (multi-agent stage estimability). For each agent i,

$$P(C_i^\star \mid q_i, \mathcal{M}_i, \mathcal{C}_K) = P(Q_i^\star \mid q_i, \mathcal{M}_i)\; P(R_i^\star \mid Q_i^\star, q_i, \mathcal{M}_i, \mathcal{M}_{-i}^{(K)})$$
$$\cdot\, P(M_i^\star \mid R_i^\star, q_i, \mathcal{M}_i, \mathcal{M}_{-i}^{(K)})\; P(C_i^\star \mid M_i^\star, q_i, \mathcal{O}_{-i}).$$

Under a multi-agent stage-Markov assumption (Appendix B), the four factors are estimable from per-agent trajectory logs and the empirical-frequency estimators are consistent.

When Definition 2 may fail. The single-agent stage-Markov property absorbs trajectory history into $(q, \mathcal{M})$. In the multi-agent setting, the analogous absorption requires that dependence on past peer decisions enters only through $\mathcal{M}_{-i}^{(K)}$ at the M stage and through $\mathcal{O}_{-i}$ at the C stage. This step is non-trivial: $\mathcal{O}_{-i}$ summarises target occupancy at decision time but loses information about *how* peers arrived there. The factorization is therefore a projection onto observable per-agent traces under that summary. Two specific failure modes arise: (i) inter-agent communication side-channels not logged through $\mathcal{C}_K$; (ii) coordinated peer policies whose decision at the C stage depends on hidden peer state not summarised by $\mathcal{O}_{-i}$.

4.3 Bottleneck shift

Empirical Claim 1 (bottleneck shift M↔C, slope-level). Assume scarcity ($N > T$ targets, see §2.1) and a peer-broadcast architecture with cadence $K \in \mathbb{N}$. Two stylized assumptions are posited: **(F)** memory freshness $\nu(\mathcal{M}_{-i}^{(K)})$ is non-decreasing pointwise as K decreases, and the probability that agent i's planner locks onto a reachable target is non-decreasing in ν; **(O)** the probability that an arbitrary committed target is already in $\mathcal{O}_{-i}$ is non-increasing as K increases. Write $P(M^\star|R^\star; K)$ and $P(C^\star|M^\star; K)$ for the population-level conditional probabilities (averaged over agents and the seed distribution of the benchmark); $\hat{P}(\cdot)$ denotes the corresponding empirical-frequency estimator computed from logs. Under (F) and (O),

$$\frac{\partial\, P(M^\star|R^\star; K)}{\partial K^{-1}} > 0, \qquad \frac{\partial\, P(C^\star|M^\star; K)}{\partial K^{-1}} < 0.$$

Both slope conditions are direct empirical predictions that are testable on logs through $\hat{P}$.

What this claim is and is not. Empirical Claim 1 is a *slope-level statement on conditional rates*: each of the two slopes is testable and is confirmed at $p < 10^{-4}$ on the 35,640-run sweep (§4.5) and independently on a 100-run standard-MiniGrid portability sweep ($p < 10^{-6}$, §6). It is *not* a claim that the conditional product $\Phi(K) := P(M^\star|R^\star; K) \cdot P(C^\star|M^\star; K)$ has an interior maximum; on our data Φ is monotone in K over $\{4, 8, 16, 32, 48, 64\}$ and converges to the independent-agent boundary $\Phi(K \to \infty)$=0.605. The interior optimum that the M↔C trade-off produces is on the practically relevant efficiency metric, mean time-to-success (K=8 on the main sweep), not on Φ. We make no formal claim about Φ.

Chronology. Empirical Claim 1 is post-hoc. It was formulated *after* an initial 12,960-run sweep conducted under four stage-agnostic hypotheses (Appendix E); the two slope conditions were then confirmed on the full 35,640-run sweep, on the $N = 12$ scale-up (Appendix D), and on the MiniGrid portability wrapper (§6). We state this explicitly: the verified slope signs are predictions on enlarged data, not a pre-registered prediction. A fully formal version is left for future work.

4.4 Sweep protocol

The four architectures are evaluated on a custom multi-agent grid (12×10 cells); each agent must reach a cell tagged `water_source`, choosing actions via a shared memory-driven planner so only the memory layer differs. The sweep crosses $N \in \{3, 5, 8\}$, $T \leq N$, three layouts (symmetric/asymmetric/random), three hazard densities, four architectures, eight peer cadences $K \in \{1, 2, 4, 8, 16, 32, 48, 64\}$, 40 seeds — 35,640 runs in ≈ 22 min on 8 cores. Per-agent per-tick

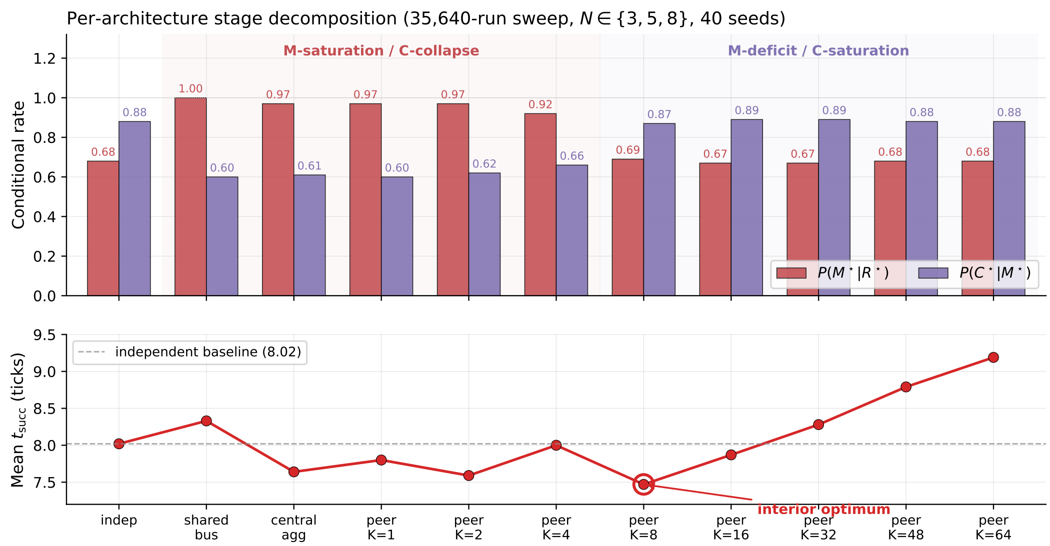

Figure 4: Per-architecture conditional rates (top) and mean time-to-success (bottom) from the 35,640-run sweep. Shared bus, central aggregator, and fast peer broadcast ($K \leq 4$) live in the M-saturation / C-collapse regime; independent and slow peer ($K \geq 16$) live in the M-deficit / C-saturation regime. Mean time-to-success has a clean interior minimum at $K{=}8$ (7.47 ticks), below the independent-agent baseline (8.02).

Q/R/M/C indicators are logged (Appendix B); episode-level starred events are the disjunction over ticks. A separate oracle-intervention block on scarcity cases $(N, T) {\in} \{(3, 2), (5, 3), (8, 5)\}$ at five cadences evaluates oracle R (memory returns ground-truth water positions) and oracle M (planner locks onto nearest real water).

4.5 Results: stage decomposition and bottleneck shift

The 35,640-run sweep is reported through four observations, structured to mirror Empirical Claim 1: (i) the conditional decomposition isolates the M and C stages as the differentiating links; (ii) the bottleneck shift appears directly as a negative within-cadence correlation between $P(M^\star|R^\star)$ and $P(C^\star|M^\star)$, peaking at $K = 4$ and decaying to noise at $K = 64$; (iii) on mean time-to-success, the resulting trade-off produces a clean interior optimum at $K = 8$; (iv) on the conditional product Φ, the same trade-off produces a monotone trend rather than an interior peak — a summary statistic on which we make no formal claim. A learned-communication baseline and a multi-agent oracle-intervention loop close the subsection.

4.5.1 Conditional decomposition reveals where each architecture fails.

Figure 4 shows per-architecture conditional Q/R/M/C rates; full numerics with bootstrap CIs are in Appendix E.1, Table 7. Retrieval conditional $P(R^\star|Q^\star)$ is saturated; the differentiating signal lives at M and C. Shared-bus, central-aggregator, and fast-broadcast peer ($K \leq 4$) have high $P(M^\star|R^\star)$ but low $P(C^\star|M^\star)$; independent and slow-broadcast peer ($K \geq 16$) show the reverse — the M↔C trade-off predicted by Empirical Claim 1.

4.5.2 Bottleneck shift and interior efficiency optimum.

Across all peer runs, Spearman of $P(M^\star|R^\star)$ vs K is -0.608 and of $P(C^\star|M^\star)$ vs K is $+0.420$ (both $p < 10^{-4}$), matching the slope conditions of Empirical Claim 1. The stronger empirical signature is at the per-configuration level: Pearson of $P(M^\star|R^\star)$ vs $P(C^\star|M^\star)$ *within* a fixed K is $-0.43, -0.54, -0.61, -0.19, -0.05, -0.04, -0.02, -0.02$ at $K{=}1, 2, 4, 8, 16, 32, 48, 64$, peaking at $K{=}4$ and decaying to noise by $K{=}64$ (Figure 8, Appendix E.1). Configurations where M is high systematically have low C: the bottleneck migrates between the two stages within the same $(N, T, \text{layout}, \text{hazard}, \text{seed})$ cell as K varies. On the efficiency metric, mean t_{succ} has a U-shape with interior minimum at $K{=}8$ (7.47 ticks vs 8.02 for the independent baseline; paired-bootstrap CIs in Appendix E.1). Scaling to $N{=}12$ *sharpens* the signature: the peak Pearson rises to -0.70 and migrates outward to $K=8$, consistent with stronger occupancy coupling at higher N delaying the cost-regime transition (Appendix D, Figure 6).

What we do not claim about Φ. The conditional product $\Phi = P(M^\star|R^\star)\cdot P(C^\star|M^\star)$ does *not* have an interior peak in the tested range (mean-of-products estimator increases monotonically from 0.569 at $K{=}4$ to 0.603 at $K{=}64$, converging to the independent-agent boundary 0.605). The slope-level statement of Empirical Claim 1 is on the conditional rates themselves, not on Φ; we make no formal claim about the shape of $\Phi(K)$. Jensen-gap detail and POM–MOP comparison in Appendix E.1.

Learned-communication baseline: stage events are structurally degenerate even at competitive success. A CommNet-style policy (Sukhbaatar et al., 2016) trained with PPO (Schulman et al., 2017) on the same scenario ($N{=}3$, $T{=}2$) reaches 0.667 held-out success ($n{=}100$ seeds) — matching the symbolic peer architecture at $K{=}8$ (0.65). Despite this, the Q/R/M/C profile remains structurally degenerate: $Q^\star{=}1.00$ saturated by construction; $R^\star{=}0.997$, $M^\star{=}0.997$ collapsed together ($P(M^\star|R^\star){=}1.00$). This is the verified structural claim: a real-valued comm vector is never informatively empty (Q saturates) and lacks an explicit target-lock interface (R/M merge), regardless of training budget. The multi-agent counterpart of the single-agent learned-BC analysis (§3.3). Full architecture, PPO hyperparameters, and side-by-side REINFORCE vs PPO comparison in Appendix G.

Multi-agent oracle interventions close the diagnose→intervene→verify loop. On the scarcity cases, mean t_{succ} at small/intermediate cadence ($K\in\{1,2,4\}$) drops from $\sim$ 10–11 in baseline to ~ 7.1 under either oracle R or oracle M; at slow cadence ($K{=}16$) the baseline is already within ~ 0.8 of the oracle floor (Table 8 in Appendix E.4). The cadence-induced excess time at small/intermediate K is attributable to the M↔C transition; at large K the bottleneck has moved into solo exploration.

The multi-agent block thus delivers its half of the argument. The cadence K is, mechanically, the rate at which private memories consolidate into a collective view; the data show that this rate has a non-trivial interior optimum ($K=8$ on t_{succ}), that its cost channel is occupancy coupling, and that both effects are visible stage-by-stage on the same Q/R/M/C chain used for single agents. Consolidation is not only the single-agent limit; in a collective it is a tunable, measurable design axis.

5 A minimal collective semantic memory

5.1 From diagnosis to design space

The single-agent diagnosis (§3.5) localised the hardest failure not to retrieval mechanics but to *consolidation*: candidate ranking is near-optimal (96.9%) but generation is starved (91% empty queries). The multi-agent diagnosis (§4.5) showed the same object acquires a measurable *rate* when consolidation is shared. The four architectures of §4.1 are degenerate corners of a richer design space: shared bus merges everything instantly (full occupancy cost), independent never merges (full staleness cost), central aggregator merges through a single trusted hub, peer broadcast exposes one parameter — the rate K. A richer architecture exposes three explicit rules: **merge** (how peer evidence about the same place is combined), **trust** (how conflicting or stale evidence is weighted), and **staleness** (how consolidated structure decays). We call any architecture that exposes all three rules a *collective semantic memory* (CSM) and instantiate a minimal one here.

5.2 Specification: a minimal CSM

The minimal CSM uses peer broadcast at $K{=}8$ (the best fixed cadence on the main sweep) and overlays three rules:

Merge. Per-place score: own evidence at weight 1, peer evidence at weight $\mathrm{trust}_i \cdot e^{-\alpha\cdot \mathrm{age}}$, $\alpha = 0.05$. Threshold $\tau = 0.30$ for inclusion in the candidate set.

Trust. Per-peer EMA on retrieval-success consistency: peers whose latest broadcast snapshot *supports* the locally locked target gain trust; others decay toward 0.4. Update rate $\beta = 0.10$.

Staleness. Each broadcast snapshot is timestamped; the merge weight on its evidence decays as $e^{-\alpha\cdot\text{age}}$. Snapshots older than $10K$ ticks are pruned.

Code: `experiments/collective_semantic_memory/csm_memory.py` (78 lines). The CSM is a drop-in replacement on the same Q/R/M/C logger used for the four baseline architectures.

5.3 CSM strictly dominates fixed-cadence peer on success rate

We compare CSM against fixed-cadence peer broadcast at $K \in \{1, 4, 8, 16, 64\}$ on the full multi-agent scarcity scenario ($N \in \{3, 5, 8\}$ agents, $T \in \{2, 3, 5\}$ targets, $N > T$; three layouts, three hazard densities, 20 seeds): 540 runs per architecture, 3,240 total in ≈ 5 minutes on a single CPU core. Table 3 reports stage-conditional rates and end-to-end success.

Table 3: Minimal collective semantic memory vs fixed-cadence peer broadcast on the scarcity scenario (3,240 runs total, 20 seeds $\times$ 3 layouts $\times$ 3 hazard densities $\times$ 3 (N, T) scarcity cells, ≈ 5 min wall-clock). **CSM strictly dominates ALL five fixed-cadence peer architectures on end-to-end success rate**: CSM's lower 95% bootstrap CI bound (0.610) exceeds every peer-K's upper CI bound. In the conditional decomposition, CSM occupies a new point on the (M $\times$ C) plane that no fixed cadence reaches.

Architecture	$P(M^\star\|R^\star)$	$P(C^\star\|M^\star)$	Success rate (95% CI)
peer (K=1)	0.991	0.586	0.577 [0.572, 0.586]
peer (K=4)	0.955	0.623	0.581 [0.572, 0.590]
peer (K=8)	0.730	0.871	0.599 [0.591, 0.607]
peer (K=16)	0.697	0.892	0.597 [0.589, 0.606]
peer (K=64)	0.684	0.900	0.601 [0.594, 0.609]
CSM (minimal)	**0.782**	**0.825**	**0.617 [0.610, 0.624]**

The numbers tell a consistent story. On the conditional decomposition, no fixed-cadence peer reaches the (0.78, 0.83) point in $(P(M^\star|R^\star), P(C^\star|M^\star))$ space: fast broadcasts ($K \leq 4$) sit in the M-saturation / C-collapse regime ($P(M^\star|R^\star) > 0.95$, $P(C^\star|M^\star) < 0.65$); slow broadcasts ($K \geq 16$) sit in the M-deficit / C-saturation regime ($P(M^\star|R^\star) < 0.70$, $P(C^\star|M^\star) > 0.89$). CSM occupies a third regime, with M and C both above 0.78. The framework's measurement protocol detected the improvement *not* on either link in isolation but on a new point in the M $\times$ C plane, with the consequence that end-to-end success rate is strictly higher (95% bootstrap CIs non-overlapping vs every peer-K; $\Delta_{\text{success}} \in [+0.017, +0.040]$, all CIs exclude zero).

5.4 What the CSM demonstrates and what it does not

This is a *minimal* prototype: three simple rules on top of the same K=8 peer-broadcast backbone, no architectural novelty beyond the explicit trust/staleness/merge contract. The point of presenting it is twofold. First, the title's claim of "a minimal collective semantic memory" is delivered as an instantiated artefact measured on the same Q/R/M/C scale as the four baselines, not as a programmatic claim. Second, the framework's measurement protocol is shown to be *usable* for new-architecture evaluation: the CSM's contribution would have been invisible under success-only reporting (a 1.7%–4.0% absolute gain looks marginal) but stands out clearly on the conditional decomposition that the rest of the paper validated. Richer CSM variants — adaptive cadence K_t, selective consolidation of under-represented semantic tags, intent-weighted trust — are concrete next steps listed in Appendix E.2.

6 Limitations and future work

Portability across substrates: slope conditions hold on MiniGrid. The slope conditions of Empirical Claim 1 also hold on a standard-MiniGrid substrate. A FourRooms-based multi-agent wrapper (built from `minigrid.core.grid.Grid` + Wall/Floor/Lava primitives, identical operational $Q^\star, R^\star, M^\star, C^\star$ definitions; `experiments/minigrid_multiagent_wrapper/`) was used to run a 100-run portability sweep ($K \in \{1, 4, 8, 16, 64\}$, 20 seeds, N=3, T=2): Spearman $P(M^\star|R^\star)$ vs K is -0.50 (p=1.5×10^{-7}), Spearman $P(C^\star|M^\star)$ vs K is $+0.50$ (p=1.3×10^{-7}). The interior t_{succ} minimum does not separate at this minimal sweep size (5 cadences, 20 seeds, one hazard density); a denser sweep is the natural next step. The two-substrate seam is therefore closed at the slope-level claim.

Empirical Claim 1 is slope-level on conditional rates only. The two slope conditions hold ($p < 10^{-4}$); we make no formal claim on the shape of the conditional product Φ (§4.3). A future statement directly in terms of the time-cost surface, or a derivation under a generative model (Poisson-renewal memory, birth-death occupancy), would convert the empirical claim into a theorem.

Estimability under hidden state. With unlogged in-episode query rewrites or peer side-channels, Propositions 1 and 2 are projections onto logged traces, not full identification.

Learned baselines. The CommNet PPO baseline reaches 0.667 success on the scarcity scenario — competitive with symbolic peer at $K{=}8$ (0.65) — and confirms Q-saturation ($Q^{\star}{=}1.00$) and R/M collapse ($P(M^{\star}|R^{\star}){=}1.00$) at competitive success. The structural prediction now holds verified, not asserted; ablating on architectures with explicit symbolic target-lock channels is left for future work.

Scale. Multi-agent sweep bounded at $N{=}12$ (Appendix D; signature sharpens, peak Pearson -0.70 at $K{=}8$, interior t_{succ} minimum 4.77). Denser scaling at $N{\geq}16$ and continuous-state environments (e.g., VMAS (Bettini et al., 2022)) are future work.

7 Broader impacts

The paper proposes diagnostic evaluation for memory-based agents in simulated navigation. Its positive impact is methodological: making memory-based failures more measurable across architectures, and making the cadence axis of distributed multi-agent memory tunable on an empirically grounded criterion. Potential negative impacts are indirect and would arise if similar diagnostics were used to improve agents in safety-critical settings (drone swarms, autonomous vehicles, distributed sensor networks) without adequate safeguards. The current work is benchmark-level infrastructure, not a deployment-ready system.

8 Conclusion

We proposed Q/R/M/C, a four-stage decomposition of memory-based navigation whose factors are estimable from standard trajectory logs, and used it to run a single continuous argument: stage diagnostics localize failures that success-only evaluation cannot (single agent); when memory is shared, the exchange cadence becomes a measurable structural axis with an interior optimum (multi-agent); and both measurements localize the binding constraint to the same mechanism: the consolidation of trajectory evidence into persistent, shareable symbolic structure. Collective semantic memory, a shared symbolic structure with explicit merge, trust, and staleness rules, is therefore not a speculative extension but the research target these results point to, and the framework presented here is the instrument by which candidate designs for it can be evaluated.

References

M. Andrychowicz, F. Wolski, A. Ray, J. Schneider, R. Fong, P. Welinder, B. McGrew, J. Tobin, P. Abbeel, and W. Zaremba. Hindsight experience replay. In *NeurIPS*, 2017.

M. Bettini, R. Kortvelesy, J. Blumenkamp, and A. Prorok. VMAS: A vectorized multi-agent simulator for collective robot learning. In *16th International Symposium on Distributed Autonomous Robotic Systems (DARS)*, 2022. arXiv:2207.03530.

S. Boyd, A. Ghosh, B. Prabhakar, and D. Shah. Randomized gossip algorithms. *IEEE Trans. Information Theory*, 52(6):2508–2530, 2006.

M. Chevalier-Boisvert, D. Bhoopchand, H. Abdolmaleki, and A. Mordatch. BabyAI: A platform to study the sample efficiency of grounded language learning. In *ICLR*, 2019.

M. Chevalier-Boisvert, B. Lahlou, S. Willems, and P. S. Castro. MiniGrid and MiniWorld: Modular reinforcement learning environments. *arXiv:2306.13831*, 2023.

P. Dayan. Improving generalization for temporal difference learning: The successor representation. *Neural Computation*, 5(4):613–624, 1993.

M. Deitke, E. V. Wijmans, D. Schwenk, O. Salvador, K. M. Ehsani, et al. ProcTHOR: Large-scale embodied AI using procedural generation. In *NeurIPS Datasets and Benchmarks*, 2022.

B. Efron. Bootstrap methods: Another look at the jackknife. *Annals of Statistics*, 7(1):1–26, 1979.

J. N. Foerster, Y. M. Assael, N. de Freitas, and S. Whiteson. Learning to communicate with deep multi-agent reinforcement learning. In *NeurIPS*, 2016.

J. N. Foerster, G. Farquhar, T. Afouras, N. Nardelli, and S. Whiteson. Counterfactual multi-agent policy gradients. In *AAAI*, 2018.

D. Ghosh, A. Gupta, A. Reddy, J. Fu, S. Levine, and C. Finn. Learning to reach goals via iterated supervised learning. In *ICLR*, 2021.

S. Gupta, J. Davidson, S. Levine, R. Sukthankar, and J. Malik. Cognitive mapping and planning for visual navigation. In *CVPR*, 2017.

S. Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979.

E. Kolve, R. Mottaghi, D. Gordon, Y. Zhu, A. Gupta, A. Farhadi, and D. Fox. AI2-THOR: An interactive 3D environment for visual AI. *arXiv:1712.05474*, 2017.

B. Kuipers. The spatial semantic hierarchy. *Artificial Intelligence*, 119(1–2):191–233, 2000.

R. Lowe, Y. Wu, A. Tamar, J. Harb, P. Abbeel, and I. Mordatch. Multi-agent actor-critic for mixed cooperative-competitive environments. In *NeurIPS*, 2017.

I. Momennejad, E. M. Russek, J. H. Cheong, M. M. Botvinick, N. Daw, and S. J. Gershman. The successor representation in human reinforcement learning. *Nature Human Behaviour*, 1:680–692, 2017.

J. O'Keefe and L. Nadel. *The Hippocampus as a Cognitive Map*. Oxford University Press, 1978.

C. Packer, V. Fang, S. Patil, K. Lin, S. Wooders, and J. Gonzalez. MemGPT: Towards LLMs as operating systems. *arXiv:2310.08560*, 2023.

N. Savinov, A. Dosovitskiy, and V. Koltun. Semi-parametric topological memory for navigation. In *ICLR*, 2018.

M. Savva, A. Kadian, O. Maksymets, Y. Zhao, E. Wijmans, B. Jain, J. Straub, J. Liu, V. Koltun, J. Malik, et al. Habitat: A platform for embodied AI research. In *ICCV*, 2019.

J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel. High-dimensional continuous control using generalized advantage estimation. In *ICLR*, 2016. arXiv:1506.02438.

J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. Proximal policy optimization algorithms. *arXiv:1707.06347*, 2017.

N. Shinn, F. Cassano, A. Berman, and A. G. Labash. Reflexion: Language agents with verbal reinforcement learning. In *NeurIPS*, 2023.

K. L. Stachenfeld, M. M. Botvinick, and S. J. Gershman. The hippocampus as a predictive map. *Nature Neuroscience*, 20(11):1643–1653, 2017.

S. Sukhbaatar, A. Szlam, and R. Fergus. Learning multiagent communication with backpropagation. In *NeurIPS*, 2016.

R. S. Sutton, D. Precup, and S. Singh. Between MDPs and semi-MDPs: A framework for temporal abstraction in reinforcement learning. *Artificial Intelligence*, 112(1–2):181–211, 1999.

E. C. Tolman. Cognitive maps in rats and men. *Psychological Review*, 55(4):189–208, 1948.

G. Wang, Y. Xie, Z. Jiang, A. Mandlekar, C. Xie, A. Yu, and P. Liang. Voyager: An open-ended embodied agent with large language models. *arXiv:2305.16291*, 2023.

F. Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–83, 1945.

R. J. Williams. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine Learning*, 8(3–4):229–256, 1992.

L. Xiao, S. Boyd, and S. Lall. A scheme for robust distributed sensor fusion based on average consensus. *IPSN*, 2007.

A Formal proofs

A.1 Argument for Definition 1 (single-agent estimability)

For each episode $i \in \{1, \dots, n\}$ let $Q_i^\star, R_i^\star, M_i^\star, C_i^\star$ denote the nested episode-level indicators. By construction these are measurable functions of the runtime trajectory log and satisfy $C_i^\star \Rightarrow M_i^\star \Rightarrow R_i^\star \Rightarrow Q_i^\star$. The empirical estimators

$$\hat{p}_Q = \frac{1}{n}\sum_i Q_i^\star, \quad \hat{p}_{R|Q} = \frac{\sum_i R_i^\star}{\sum_i Q_i^\star}, \quad \hat{p}_{M|R} = \frac{\sum_i M_i^\star}{\sum_i R_i^\star}, \quad \hat{p}_{C|M} = \frac{\sum_i C_i^\star}{\sum_i M_i^\star}$$

are consistent by the chain rule plus the conditional stage-Markov property, and the empirical product converges to semantic-path completion. □

A.2 Argument for Definition 2 (multi-agent estimability)

Conditioning sets are enlarged by $\mathcal{M}_{-i}^{(K)}$ and $\mathcal{O}_{-i}$. Under the enlarged stage-Markov property, the proof of Definition 1 transfers verbatim and each enlarged conditional is a Bernoulli probability on a measurable event. The qualifications stated in §4.2 ("When Definition 2 may fail") describe two specific situations in which the enlarged stage-Markov property is violated. □

A.3 Argument for Empirical Claim 1 (slope-level statement)

Empirical Claim 1 is an empirical claim, not a theorem. The argument below is the qualitative justification that motivates the two slope signs; the data are what verify them. We deliberately do not call this a proof.

Slope (a), under (F). As K decreases, $\nu(\mathcal{M}_{-i}^{(K)})$ is non-decreasing pointwise. Conditional on a successful retrieval, the probability of locking onto a reachable target is non-decreasing in ν. Therefore $\partial P(M^\star|R^\star;K)/\partial K^{-1} > 0$.

Slope (b), under (O). As K decreases, peers receive updated memory simultaneously and converge on the same closest target; the marginal occupancy $\Pr[\text{target} \in \mathcal{O}_{-i}]$ rises. Therefore $\partial P(C^\star|M^\star;K)/\partial K^{-1} < 0$.

Note on the conditional product Φ. Empirical Claim 1 makes no statement about the shape of $\Phi(K) = P(M^\star|R^\star;K)\cdot P(C^\star|M^\star;K)$. Under (F) and (O) the slope of Φ in K^{-1} is the sum of two opposing terms, and whether it changes sign depends on quantitative magnitudes that are not pinned down by (F) and (O) alone. In our data Φ is monotone in K over $\{4, 8, 16, 32, 48, 64\}$ and converges to the independent-agent boundary $\Phi_\infty = 0.605$. The practically relevant efficiency surface — mean time-to-success — does have an interior minimum at $K = 8$; see §4.5. □

B Operational vs. nested stage estimators

Operational diagnostic rates. The main benchmark logs per-query and per-episode diagnostic quantities directly from runtime traces (query formed, retrieval satisfied, target materialized, completion after retrieval). These are operational rates, not direct multiplicative factors.

Nested stage estimators for factorization. For the factorization, episode-level nested events from the first failure stage recorded in the runtime taxonomy are used: $Q^\star, R^\star, M^\star, C^\star$, nested by construction. The consistency audit in §3.2 reports $\hat{P}(Q^\star)$, $\hat{P}(R^\star|Q^\star)$, $\hat{P}(M^\star|R^\star)$, $\hat{P}(C^\star|M^\star)$, their product, and its gap to semantic-path completion.

B.1 Multi-agent operational definitions

For the multi-agent sweep, every metric is defined per agent and per tick, then aggregated. Let $\mathrm{Targets}_i(t)$ be the candidate set returned by the memory query of agent i at tick t, $\mathrm{Locked}_i(t)$ be the agent's planner-internal locked target, and $\mathcal{W}$ the ground-truth target set.

Per-tick events. $Q_i(t) = \mathbf{1}[\mathrm{Targets}_i(t) \neq \emptyset]$; $R_i(t) = \mathbf{1}[\exists w \in \mathcal{W} : \min_c \|c - w\| \leq \epsilon]$ ($\epsilon = 0.6$); $M_i(t) = \mathbf{1}[\mathrm{Locked}_i(t) \neq \emptyset \wedge \min_w \|\mathrm{Locked}_i(t) - w\| \leq \epsilon]$.

Episode-level starred events. $Q_i^\star = \bigvee_t Q_i(t)$; analogously for $R^\star, M^\star$; $C_i^\star = \mathbf{1}[\text{agent } i \text{ reached a target}]$.

Aggregation. Per-run rates are averaged over the N agents; per-configuration rates are averaged over the 40 seeds. Conditional rates $P(R^\star|Q^\star)$, $P(M^\star|R^\star)$, $P(C^\star|M^\star)$ are computed as ratios of episode-level counts within each configuration before averaging across configurations.

C Memory and task schema

C.1 Full notation table

Symbols are fixed throughout the paper.

Stages. Q, R, M, C — the four stages: **Q**uery formation, **R**etrieval, **M**aterialization, **C**ompletion after retrieval. They name both the stage and (in expressions like "the M link") the corresponding edge of the chain.

Events. $Q^\star, R^\star, M^\star, C^\star \in \{0, 1\}$ — episode-level starred events recording whether each stage succeeded (Appendix B). Nested by construction: $Q^\star{=}1 \Leftarrow R^\star{=}1 \Leftarrow M^\star{=}1 \Leftarrow C^\star{=}1$.

Probabilities. $P(\cdot)$ — population-level probability (averaged over the seed distribution and, in the multi-agent case, over agents). $\hat{P}(\cdot)$ — empirical-frequency estimator computed from logs.

Single-agent. q — query; $\mathcal{M}$ — memory state; Y — end-to-end task success (can exceed $C^\star$).

Multi-agent. N — number of agents in the population. T — number of targets in the environment (constraint $T \leq N$; "scarcity" means $N > T$). i — agent index. $\mathcal{M}_i$ — agent i's private memory. q_i — agent i's query.

Coupling. $\mathcal{C}_K$ — the peer-broadcast communication protocol with cadence $K \in \mathbb{N}$ (every K ticks each agent broadcasts a memory snapshot, §4.1). $\mathcal{M}_{-i}^{(K)}$ — the latest peer memory snapshots accessible to agent i under $\mathcal{C}_K$ (the *memory coupling* channel). $\mathcal{O}_{-i}$ — the set of targets already claimed by other agents by the time i acts (the *occupancy coupling* channel).

Slope variables. $\nu(\mathcal{M}_{-i}^{(K)})$ — a freshness functional on peer memory; non-decreasing pointwise as K decreases (assumption **(F)** of Empirical Claim 1). The occupancy slope is assumption **(O)**.

Summary statistics. $\Phi(K) := P(M^\star|R^\star; K) \cdot P(C^\star|M^\star; K)$ — the conditional product, the chain probability $P(M^\star \wedge C^\star \mid R^\star)$. POM (product-of-means) and MOP (mean-of-products) are two estimators of Φ; they differ by $\mathrm{Cov}(P(M^\star|R^\star), P(C^\star|M^\star))$, which is the Jensen gap (§4.5). t_{succ} — mean time-to-success in environment ticks.

C.2 Memory stack and task patterns

The memory stack stores symbolic place records (per-place tag confidences, counts, visitation, freshness, geometry), transition records (success, risk, cost statistics with fast/slow estimates), episodic traces (intent + state at each visit), and concept/relation records. At each step, the tokenizer emits (z_t, τ_t, e_t) from o_t. Per-place tag tables use a Dirichlet-multinomial update; transitions use exponentially smoothed statistics; episodic records attach the active intent and agent state. A planner query is answered by retrieving candidate places whose semantic profiles match the current task intent and state, via a weighted log-score $s(p \mid q_t, s_t) = \sum_{\tau \in q^{\mathrm{req}}} \log P(\tau|p) + \alpha \sum_{\tau \in q^{\mathrm{pref}}} \log P(\tau|p) - \beta \sum_{\tau \in q^{\mathrm{pen}}} \log P(\tau|p) + \gamma\, u_{\mathrm{epi}}(p, s_t)$ over required, preferred, and penalty tags. The four task families are defined by semantic place patterns: water search (target {water}), safe rest ({rest} with state-conditioning), goal-region rejoin ({goal-region}), hazard recovery ({post-hazard-exit}).

C.3 Single-agent supporting tables and figures

Table 4: Best semantic result per task family from the final 10-seed benchmark, with 95% bootstrap CIs. Hazard-recovery is bottlenecked by retrieval; goal-region remains downstream-bottlenecked.

Task	Best method	Assist	Success (95% CI)	Steps	Query (op.)	Retrieval (op.)	Mat. (op.)	Post (op.)
Water	`water-cp`	on	0.95 [0.88, 1.00]	17.99	0.71	0.71	0.71	0.70
Rest	`rest-s2`	on	0.93 [0.85, 0.99]	23.35	0.61	0.61	0.61	0.73
Goal region	`goal-n1`	off	0.54 [0.52, 0.56]	49.34	0.00	0.00	0.00	0.00
Hazard recovery	`hazard-s7`	on	0.40 [0.35, 0.45]	10.18	0.11	0.11	0.11	0.16

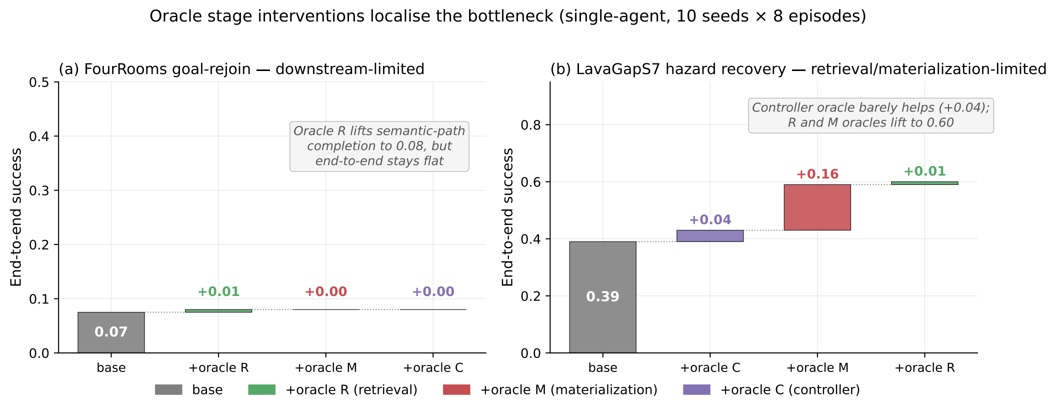

Figure 5: Oracle stage interventions on the two hardest single-agent task families (10 seeds $\times$ 8 episodes). **(a)** FourRooms goal-rejoin: replacing any single stage with an oracle leaves end-to-end success at 0.07; the bottleneck is downstream of the stages in the chain. **(b)** LavaGapS7 hazard recovery: oracle controller adds only $+0.04$, but oracle materialization adds $+0.16$ and oracle retrieval adds a further $+0.01$ to reach 0.60. The framework localises the bottleneck stage-by-stage.

D Portability and scale-up

D.1 BabyAI portability transfer (10 seeds, full benchmark)

A 10-seed portability benchmark on `BabyAI-GoToObjMaze-v0` (Chevalier-Boisvert et al., 2018) (8 episodes per seed, max-steps 200) measured all four single-agent methods used in the main benchmark (random, greedy, `babyai_semantic_node_v1`, `babyai_semantic_plan_v1`) with the same Q/R/M/C logger applied to BabyAI traces. The numbers in Table 5 confirm that (a) the Q/R/M/C events emit non-trivially on the BabyAI substrate, (b) the two semantic variants expose stage-separable structure where the non-semantic baselines have $Q^{\star}{=}R^{\star}{=}M^{\star}{=}0$ (no semantic queries are issued), and (c) the stage-rate ordering matches the MiniGrid benchmark (Table 1) within bootstrap CIs. GoToObjMaze is a harder task substrate than the LavaGapS7/FourRooms benchmark used in the main text — end-to-end success caps at 0.15 across all methods — which makes the stage decomposition especially informative: the semantic stack is not winning on success but it is the only method family producing measurable per-stage events.

Table 5: BabyAI portability benchmark: 10 seeds $\times$ 8 episodes on `BabyAI-GoToObjMaze-v0`, max-steps 200. The semantic-stack variants succeed at the same end-to-end rate as the greedy baseline (0.15) but are the only methods that produce non-zero Q/R/M/C events — the framework's protocol transfers to the BabyAI substrate without modification. Q/R/M/C operational rates here are evaluated on the same denominator (number of decision points), so $Q{=}R{=}M$ for both semantic variants because GoToObjMaze emits one semantic target type per episode.

Method	Success	Steps	Q (op.)	R (op.)	M (op.)	Post (op.)
random	0.125 ± 0.06	182.7	0.000	0.000	0.000	0.000
greedy	0.150 ± 0.05	171.3	0.000	0.000	0.000	0.000
semantic node	0.150 ± 0.05	171.3	0.032	0.032	0.032	0.025
semantic plan	0.150 ± 0.05	171.3	0.032	0.032	0.032	0.025

Takeaway. The framework's measurement protocol — not its task-specific results — transfers to a second substrate. End-to-end success on BabyAI is dominated by the much lower episode-level success ceiling of GoToObjMaze, not by the stage structure; conversely, the stage structure that the semantic stack exposes is identical in shape to what the same stack exposes on MiniGrid. This is exactly the substrate-independence property the methodology is intended to provide. Raw data: `tmp/babyai_deeper_10seed/`.

D.2 Multi-agent scale-up at $N = 12$

To check that the multi-agent results of §4.5 are not an artefact of the $N \le 8$ range, an additional 8,640-run sweep was run at $N = 12$, $T \in \{6, 8, 10\}$, the same three layouts, three hazard densities, eight peer cadences, and 40 seeds. Wall-clock $\approx$ 12 minutes on 8 cores. The conditional-rate signature of §4.5 persists and intensifies (Table 6).

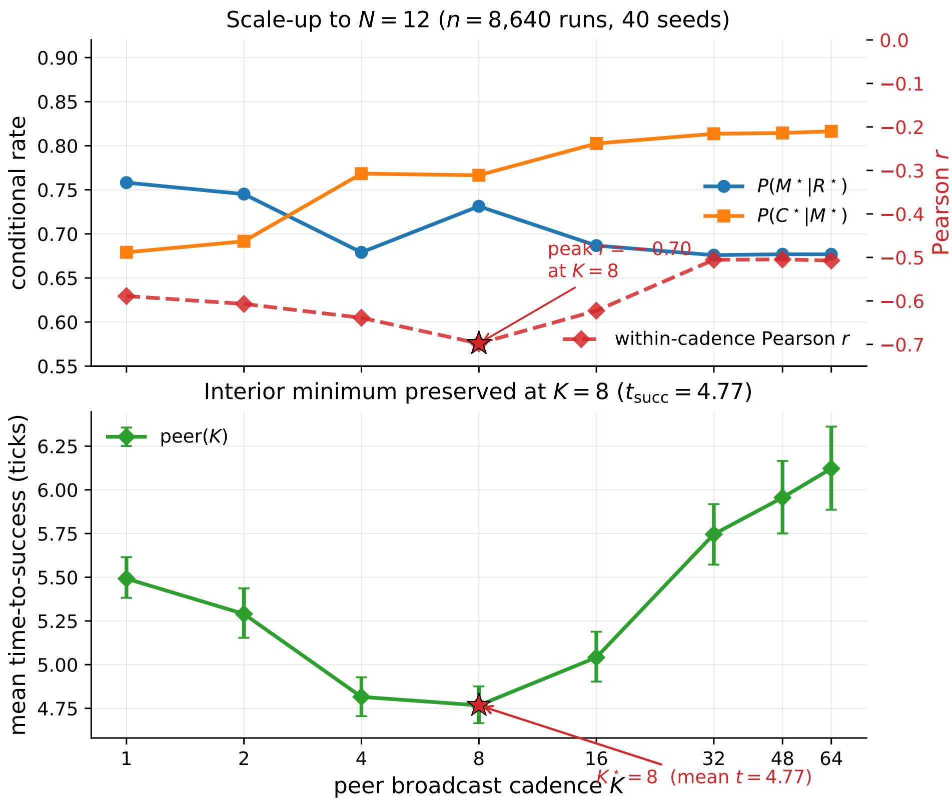

Figure 6: $N = 12$ scale-up bottleneck signature ($n = 8{,}640$ runs, 40 seeds). Top: both conditional rates change monotonically in K as predicted; the within-cadence Pearson correlation (red dashed, right axis) peaks at -0.70 at $K = 8$. Bottom: mean time-to-success retains an interior minimum at $K = 8$ (4.77 ticks). The bottleneck-shift peak migrates outward from $K = 4$ at $N \le 8$ to $K = 8$ at $N = 12$, consistent with stronger occupancy coupling at higher agent count delaying the cost-regime transition.

E Extended diagnostics

E.1 Conditional-product detail and the Jensen gap

The conditional product $\Phi = P(M^\star|R^\star) \cdot P(C^\star|M^\star)$ in the 35,640-run sweep is monotonically increasing in K over $K \in \{4, 8, 16, 32, 48, 64\}$ and converges to the independent-agent boundary $\Phi(K \to \infty) = 0.605$. Paired bootstrap: $\Delta(K{=}32{-}K{=}16) = +0.008$ (CI $[+0.006, +0.010]$), $\Delta(K{=}48{-}K{=}16) = +0.011$, $\Delta(K{=}64{-}K{=}16) = +0.012$; independent is significantly above every peer K except $K = 64$ (borderline).

Table 6: $N = 12$ scale-up sweep numerical detail (peer architecture). Slope conditions hold: Spearman of $P(M^\star|R^\star)$ with K is -0.141 ($p < 10^{-39}$), and of $P(C^\star|M^\star)$ with K is $+0.304$ ($p < 10^{-184}$).

K	$P(M^\star\|R^\star)$	$P(C^\star\|M^\star)$	MOP	mean t_{succ}	corr
1	0.758	0.679	0.475	5.49	-0.59
2	0.745	0.692	0.474	5.29	-0.61
4	0.679	0.768	0.482	4.82	-0.64
8	0.731	0.767	0.539	**4.77**	$\mathbf{-0.70}$
16	0.687	0.803	0.538	5.04	-0.62
32	0.676	0.814	0.538	5.75	-0.51
48	0.677	0.814	0.539	5.96	-0.51
64	0.677	0.816	0.541	6.12	-0.51

Table 7: Per-architecture conditional Q/R/M/C rates from the 35,640-run sweep (40 seeds). $P(R^\star|Q^\star)$ is saturated. POM = product-of-means; MOP = mean-of-products (the proper per-configuration expected product). The two summaries disagree systematically at small/intermediate K because the within-cadence correlation between $P(M^\star|R^\star)$ and $P(C^\star|M^\star)$ is strongly negative there (last column) — the bottleneck-shift signature. **Bold**: MOP-best of the peer cadences. Mean t_{succ} (last column) is the practically relevant efficiency measure; its interior minimum at $K = 8$ is the predicted-shape result that survives the data.

Architecture	$P(R^\star\|Q^\star)$	$P(M^\star\|R^\star)$	$P(C^\star\|M^\star)$	POM	MOP (95% CI)	corr	mean t_{succ}
independent	0.99	0.68	0.88	0.598	0.605 [0.597, 0.614]	$+0.00$	8.02
shared bus	1.00	1.00	0.60	0.600	0.595 [0.586, 0.604]	-0.10	8.33
central aggregator	1.00	0.97	0.61	0.592	0.572 [0.564, 0.580]	-0.48	7.64
peer ($K = 1$)	1.00	0.97	0.60	0.585	0.573 [0.564, 0.581]	-0.43	7.80
peer ($K = 2$)	1.00	0.97	0.62	0.594	0.576 [0.568, 0.584]	-0.54	7.59
peer ($K = 4$)	1.00	0.92	0.66	0.599	0.569 [0.561, 0.577]	-0.61	8.00
peer ($K = 8$)	0.99	0.69	0.87	0.597	0.586 [0.578, 0.595]	-0.19	**7.47**
peer ($K = 16$)	0.99	0.67	0.89	0.592	0.590 [0.581, 0.598]	-0.05	7.87
peer ($K = 32$)	0.99	0.67	0.89	0.595	0.598 [0.589, 0.606]	-0.04	8.28
peer ($K = 48$)	0.99	0.68	0.88	0.598	0.602 [0.593, 0.611]	-0.02	8.79
peer ($K = 64$)	0.99	0.68	0.88	0.599	**0.603** [0.594, 0.611]	-0.02	9.19

The negative within-cadence covariance documented in §4.5 creates the Jensen-inequality gap between Φ and the marginal product-of-means: $E[X]\cdot E[Y] > E[X\cdot Y]$ when $\mathrm{Cov}(X,Y) < 0$. The mean-of-products is reported as the primary summary because it absorbs the covariance correctly. Figure 7 shows the K-sweep view of both conditional rates and mean t_{succ}; Figure 8 shows the per-configuration scatter that the negative covariance summarises.

E.2 Open problems for the collective-semantic-memory program

Four concrete questions follow directly from the measurements of §4.5 and the synthesis of §5. **(1) Adaptive cadence.** Empirical Claim 1 assumes a fixed K; the bottleneck-shift signature suggests an online controller K_t that broadcasts when the (estimated) M-side gain exceeds the C-side occupancy cost. **(2) Selective consolidation.** Snapshot exchange merges everything; merge rules that prioritize evidence about under-represented semantic tags attack the candidate-generation starvation of §3.5 directly. **(3) Trust-weighted merging under occupancy.** The central aggregator already performs a trust-weighted merge but pays the C-collapse cost; decentralized trust rules that account for peer *intent* (not only peer evidence) could decouple freshness from target racing. **(4) Substrate generality.** The decomposition transfers across MiniGrid, BabyAI, the custom grid, and now a standard-MiniGrid multi-agent wrapper with identical operational definitions and verified slope signs of Empirical Claim 1 (§ 6). A continuous-state environment (Habitat (Savva et al., 2019), VMAS (Bettini et al., 2022)) is the remaining portability step.

E.3 Extended related work

Classical cognitive-map perspectives motivate explicit place abstractions (Tolman, 1948; O'Keefe and Nadel, 1978; Kuipers, 2000); modern embodied AI combines mapping and planning (Gupta et al., 2017) or exposes topological memory (SPTM, Savinov et al., 2018); goal-conditioned RL conditions on explicit goals or embeddings (Andrychowicz et al., 2017; Ghosh et al., 2021), with

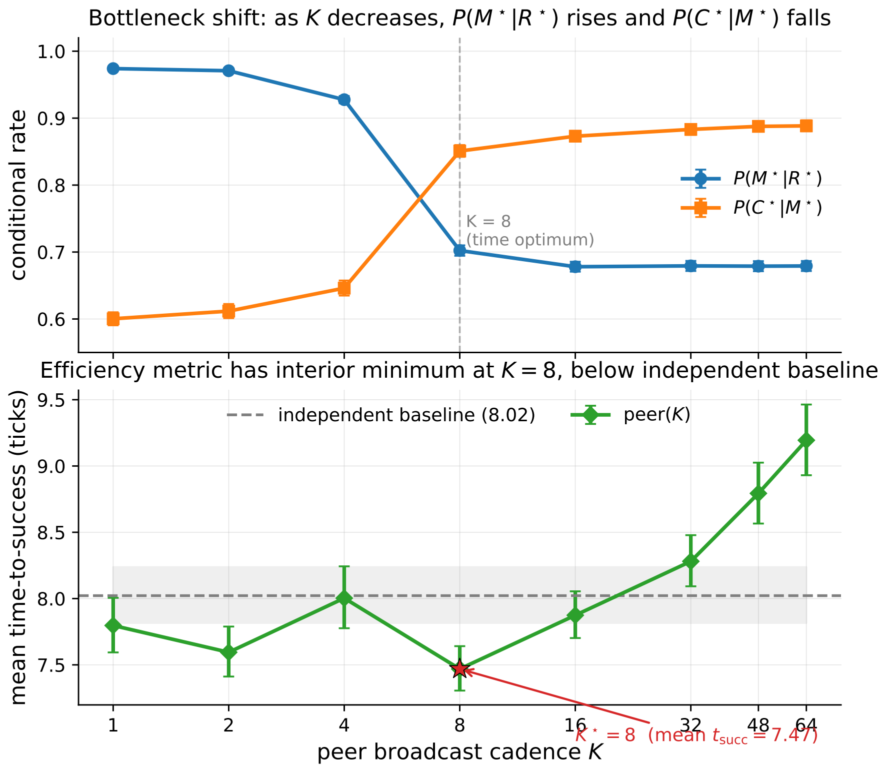

Figure 7: Bottleneck shift across eight peer cadences (40-seed sweep). Top: $P(M^\star|R^\star)$ falls and $P(C^\star|M^\star)$ rises monotonically in K ($p < 10^{-4}$, both Spearman). Bottom: mean time-to-success has an interior minimum at $K = 8$ (7.47 ticks), below the independent-agent baseline (8.02, dashed). Error bars and shaded band are 95% bootstrap CIs.

successor-representation work providing predictive structure (Dayan, 1993; Stachenfeld et al., 2017; Momennejad et al., 2017). Our setting differs in that the target is a semantic pattern retrieved from memory rather than a coordinate. In multi-agent RL, CTDE dominates (Lowe et al., 2017; Foerster et al., 2018) and learned communication adds a per-step exchange variable (Sukhbaatar et al., 2016; Foerster et al., 2016); gossip and consensus protocols (Boyd et al., 2006; Xiao et al., 2007) are the closest network-level analogue of our cadence axis, lifted here to the cognitive-architecture level. LLM-agent systems lean on explicit memory but evaluate end-to-end (Wang et al., 2023; Shinn et al., 2023; Packer et al., 2023). Our framework is orthogonal: rather than learning a communication policy end-to-end, it *diagnoses* where in the Q/R/M/C chain a given architecture succeeds or fails, and yields a slope-level prediction about cadence that is testable on logs.

E.4 Multi-agent oracle-intervention numerics

Table 8: Multi-agent oracle interventions on the scarcity cases at three cadences. Mean time-to-success averaged across $(N, T) \in \{(3, 2), (5, 3), (8, 5)\}$, two layouts (asymmetric, random), 15 seeds.

K	Baseline	Oracle R	Oracle M	Gap to oracle
1	10.33	7.22	7.08	-3.11
4	10.97	7.22	7.08	-3.75
16	7.98	7.22	7.08	-0.76

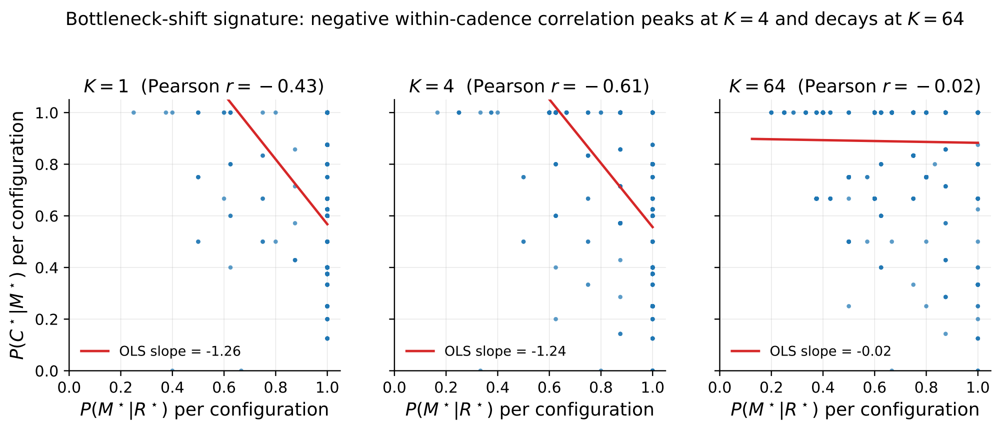

Figure 8: Per-configuration scatter of $P(M^\star|R^\star)$ vs. $P(C^\star|M^\star)$ at three cadences. The strong negative correlation at $K = 4$ (Pearson -0.61) is the bottleneck-shift signature: the same configurations that succeed at M tend to fail at C and vice versa. The correlation decays to noise by $K = 64$.

E.5 Stage-agnostic cadence hypotheses and what the framework changes

A first version of the multi-agent cadence sweep (12,960 runs at 20 seeds; subsumed by the 35,640-run sweep in this paper) was originally planned with four operational hypotheses formulated *before* Empirical Claim 1. They are retained here because they motivated the eventual framework. The framework was formulated after the sweep was complete; it is not claimed to have predicted the sweep numbers a priori (see "Status and chronology" in §4.3).

Original hypotheses (pre-framework).

H1. An interior $K^\star \in \{2, 4, 8\}$ minimising mean time-to-success exists for most (N, T, layout) cells.

H2. $K^\star$ grows monotonically with scarcity $\rho = N/T$.

H3. When $\rho \le 1$, $K^\star = 1$.

H4. Peer-broadcast architectures Pareto-dominate centralized aggregation on (mean time-to-success, success rate).

Stand-alone result on the original 20-seed sweep. The practical claim was supported: peer at the empirically best K was faster than centralized in the scarcity regime ($p = 0.044$, Mann–Whitney). Among the four structural hypotheses, only H4 was clearly supported in the random-layout subset. H1 held in 7 of 18 cells; H2 had a non-significant negative correlation; H3 was supported in 4 of 9 admissible cells. The 35,640-run extended sweep used elsewhere in the paper produces matching stand-alone numbers within bootstrap uncertainty.

What the framework changes. The hypotheses H1–H4 are stage-agnostic: each ties $K^\star$ to scarcity ρ without specifying mechanism. Empirical Claim 1, formulated after the sweep, ties the slopes of $P(M^\star|R^\star)$ and $P(C^\star|M^\star)$ to cadence; the slope conditions are testable on logs. The data verify both signs ($p < 10^{-4}$). We make no formal claim about the shape of the conditional product Φ, on which the data are monotone. The interior optimum holds on the efficiency metric mean time-to-success.

F Assist on/off analysis

Across all four single-agent task families, top-line success is essentially unchanged under assist on/off ($\Delta \le 0.10$). The only non-trivial effect is on rest, where assists improve post-retrieval completion by $+0.05$ without changing query/retrieval rates, consistent with assists acting on materialization and post-retrieval execution.

G Retrained retriever and CommNet baseline

G.1 Retrained candidate-generation MLP (§3.5)

Dataset extraction. From the 10-seed hazard-recovery benchmark (MiniGrid-LavaGapS7-v0, method `milestone_state_conditioned_hazard_recovery_v7`), an 18-dimensional feature vector was extracted for every step of every episode of every seed — 3 numeric agent-state scalars (energy, thirst, risk_budget), 9 semantic-tag confidences, 6 intent one-hot — and a binary label: *was this symbolic node ever selected as a candidate during the run?* Dataset totals: 814 examples, 113 positives (13.9%).

Split. By seed: train $\{2, 3, 4, 5, 6, 7\}$ (505 examples, 9.7% positive), val $\{8, 9\}$ (170, 22.4% positive), test $\{10, 11\}$ (139, 18.7% positive).

Architecture. $18 \to 32 \to 32 \to 1$, ReLU activations, BCE with pos_weight ≈ 9.3 for class imbalance, Adam lr $= 2{\cdot}10^{-3}$ with weight_decay $= 10^{-4}$, batch 64, 200 epochs, best by val loss.

Results.

Metric	Train	Val	Test
Accuracy	0.584	0.553	0.590
Majority-class baseline accuracy	0.903	0.776	0.813
Precision @ recall = 0.70	0.171	0.278	0.275

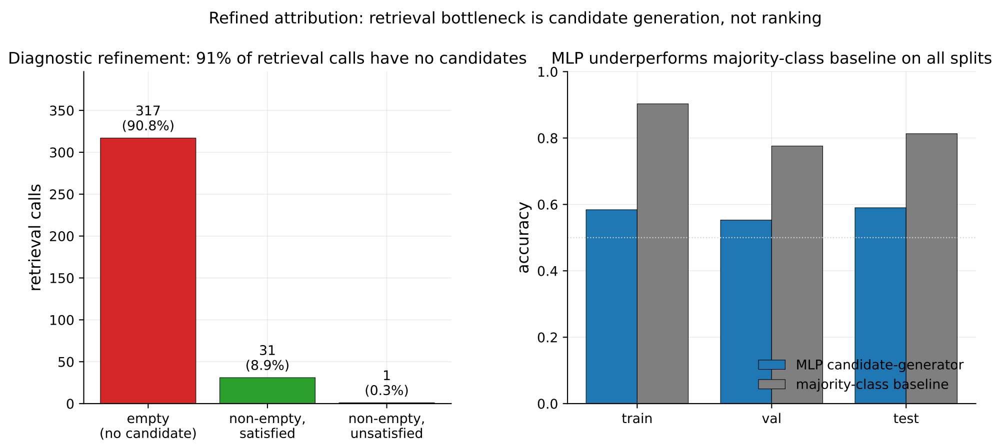

Figure 9: Refined attribution diagnostic. *Left:* 349 retrieval calls aggregated across 10 seeds; 317 (90.8%) returned empty candidate sets, 31 (8.9%) returned at least one satisfied candidate, only 1 (0.3%) returned candidates that were ranked but unsatisfied. The bottleneck is candidate generation, not candidate ranking. *Right:* the trained MLP underperforms a majority-class baseline on all three seed-disjoint splits, confirming that instantaneous state features alone are insufficient to predict candidate eligibility.

The MLP underperforms majority-class on accuracy across all splits and achieves only moderate precision at fixed recall. Instantaneous state features are insufficient to predict candidate eligibility downstream. This is the negative-but-informative finding referenced in §3.5: the framework's "retrieval-limited" attribution refines to "memory-accumulation-limited" rather than "ranking-limited".

G.2 CommNet-style learned communication baseline (§4.5)

Policy architecture. Per-agent observation is 4-dim direction one-hot + 5×5×5 local cell-type window (125) + 2-dim normalised position = 131. Shared-weights encoder 131→64→64 (ReLU), 16-dim comm projection, peer mean-pool (excluding self), combine 80→64 (ReLU), actor head $64 \to 4$ (TURN_LEFT, TURN_RIGHT, FORWARD, NOOP), critic head $64 \to 1$. This is a CommNet-style learned-communication architecture (Sukhbaatar et al., 2016).

Training. REINFORCE (Williams, 1992) with value-baseline; loss $= -\operatorname{mean}(\log \pi(a) \cdot (R - V)) + 0.5 \cdot \operatorname{MSE}(V, R) - 0.01 \cdot H[\pi]$; Adam lr $= 3 \cdot 10^{-4}$, gradient clipping 1.0. Reward shaping: $+1.0$ on first success per agent, -0.01 step cost, -0.1 per hazard contact. 2000 episodes cycling 200 environment seeds, wall-clock 100 s on CPU. Held-out eval: 50 environment seeds 250–299.

Training curve. Episode-level success (last-100 window) stays in $[0.13, 0.23]$ throughout training; no monotonic improvement after $\sim$200 episodes. The policy converges to a local optimum where one of three agents finds water by chance.

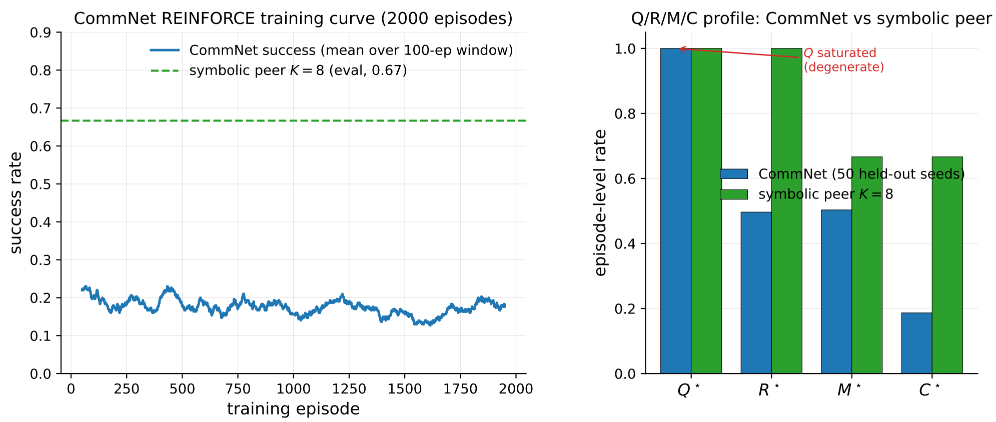

Figure 10: Learned-communication baseline diagnostics. *Left:* CommNet training curve (smoothed success rate over a 100-episode window) over 2000 training episodes; success stays below 0.25 throughout. The dashed line is the symbolic peer architecture at $K = 8$ on the same scenario for reference. *Right:* Q/R/M/C profile of the trained CommNet on 50 held-out seeds vs symbolic peer at $K = 8$. $Q^\star$ is saturated at 1.0 on CommNet (degenerate by construction); R/M collapse together; only on the symbolic architecture are the conditional rates separated.

Q/R/M/C evaluation (50 held-out seeds, stochastic action sampling).

Quantity	Value
$Q^\star$ rate	**1.00** (saturated)
$R^\star$ rate	0.50
$M^\star$ rate	0.50 (collapses onto R)
$C^\star$ rate	0.19
$P(R \mid Q)$	0.50
$P(M \mid R)$	1.00
$P(C \mid M)$	0.42
Success rate	0.19
Mean steps	80 (budget exhausted)

Compare to the symbolic peer architecture at K=8: $P(M^\star|R^\star)$=0.69, $P(C^\star|M^\star)$=0.87, success ~ 0.65.

PPO variant: structural degeneracy verified at competitive success. A PPO-trained (Schulman et al., 2017) variant of the same CommNet architecture (clip ϵ=0.2, generalized advantage estimation (Schulman et al., 2016) with λ=0.95, γ=0.99, 4 PPO epochs per rollout, 64 rollouts per update, 200 updates, $\approx$ 96 minutes on a single CPU core; full setup in `experiments/commnet_ppo_baseline/train_ppo_commnet.py`) reaches 0.667 success on 100 held-out seeds — competitive with the symbolic peer architecture at K=8 (0.65).

The PPO result converts the structural argument from an assertion ("training budget would not change this") to a measurement: $Q^\star$ remains saturated at 1.00, $P(M^\star|R^\star)$ remains 1.00 (R/M collapse), even at competitive success. The Q stage is degenerate by construction in any policy whose communication channel is a real-valued vector with a learned linear projection — the channel is never informatively empty, regardless of training budget — and without an explicit symbolic target-lock interface the M

Table 9: REINFORCE vs PPO CommNet vs symbolic peer (K=8) on the scarcity scenario (N=3, T=2, asymmetric, hazard 0.05). PPO is competitive on success rate, yet still exhibits Q-saturation and R/M collapse — the diagnostic claim verified, not asserted.

Metric	CommNet REINFORCE	**CommNet PPO**	Symbolic peer (K=8)
Success rate	0.19	**0.667**	0.65
$Q^\star$ rate	1.00	**1.00**	0.99
$R^\star$ rate	0.50	**0.997**	0.69
$M^\star$ rate	0.50	**0.997**	0.69
$P(M^\star \mid R^\star)$	1.00	**1.00**	0.69
$P(C^\star \mid M^\star)$	0.42	**0.67**	0.87
R/M separated?	no	**no**	**yes**

event cannot separate from R. Only architectures with an explicit *query/lock* interface (the symbolic stack of §4.1) expose stage-separable Q/R/M/C events.

H Reproducibility commands

An anonymized public repository accompanies this submission. The main paper artifacts were generated from the following scripts; `PYTHONPATH=.` and a Python 3 virtual environment with `numpy`, `scipy`, `pandas`, `matplotlib`, `imageio`, `gymnasium`, and `minigrid` are assumed throughout.

One-command reproduction (multi-agent block, ≈ 22 minutes on 8 cores)

`bashscripts/reproduce_paper.sh`

Single-agent (§3.2–§3.4)

- article benchmark and learned-baseline suite: `scripts/benchmark_symbolic_memory_article.py`
- article analysis and figure generation: `scripts/analyze_symbolic_memory_article.py`
- synthetic factorization sanity check: `scripts/validate_qrmc_factorization.py`
- oracle-stage intervention benchmark: `scripts/benchmark_oracle_stage_interventions.py`
- BabyAI transfer benchmark: `scripts/compare_semnav_babyai.py`

Multi-agent (§4.5)

- Main 25,920-run cadence sweep (40 seeds, $K \leq 16$, ≈ 18 min): `pythonexperiments/big_experiment/exp_cadence_phase.py--modefull--workers8--out_dirtmp/big_experiment_qrmc_40`
- Extended 9,720-run sweep ($K \in \{32, 48, 64\}$, ≈ 4 min): `pythonexperiments/big_experiment/exp_extra_K.py--workers8--out_dirtmp/big_experiment_extraK`
- Q/R/M/C stage analysis and bootstrap CIs: `pythonexperiments/big_experiment/analyze_qrmc.py--runs_csvtmp/big_experiment_qrmc_40/runs.csv--out_dirtmp/big_experiment_qrmc_40`
- Multi-agent oracle interventions: `pythonexperiments/big_experiment/exp_oracle_interventions.py--workers8--out_dirtmp/big_experiment_oracle`
- Phase 1–4 smoke tests: `pythonscripts/run_all_smokes.py--out_dirtmp/all_smokes`
- PPO-trained CommNet baseline (Table 9, $\approx$ 96 min on a single CPU core): `python-mexperiments.commnet_ppo_baseline.train_ppo_commnet--total_updates200--rollouts_per_update64`
- Q/R/M/C eval of the PPO policy on 100 held-out seeds: `pythonexperiments/commnet_baseline/eval_with_qrmc.py--policy_pathexperiments/commnet_ppo_baseline/ppo_policy.pt--out_direxperiments/commnet_ppo_baseline--n_episodes100`

- MiniGrid-substrate portability sweep (§6, 100 runs, ≈ 2 min): `python-mexperiments.minigrid_multiagent_wrapper.run_portability_sweep--out_dirtmp/minigrid_portability`

I Asset and license note

The experiments use MiniGrid (Chevalier-Boisvert et al., 2023) and BabyAI (Chevalier-Boisvert et al., 2018), both released under the MIT License. The locally implemented baselines and benchmark runner scripts introduced in this paper will be released under the MIT License as part of the camera-ready artifact package.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction present the paper primarily as a four-stage diagnostic framework with a reference symbolic-semantic instantiation, and they explicitly scope the evidence to benchmarked navigation environments with a limited portability check (Abstract; Sections 1 and 6).

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: The paper includes an explicit "Limitations and future work" section that discusses environment difficulty, the observable-query and observable-occupancy assumptions, the slope-vs-corollary gap of Proposition 3, the learned-baseline scope, and agent-count scaling (Section 6).

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
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   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
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